8) Kararlılık : $\int_{-\infty}^{\infty} |h(\tau)|\,d\tau < \infty$

$h(t) = 3^t u(-t+10) \longrightarrow \int_{-\infty}^{10} 3^t dt = \frac{3^t}{\log 3}\Big|_{-\infty}^{10} = \frac{1}{\log 3}\left(3^{10} - 3^{-\infty}\right) = 3^{10}/\log 3$ ← $\log(x) = \ln(x)$ Natural logarithm

Kararlı

$h(t) = e^{-(1-2j)t}u(t) \longrightarrow \int_{0}^{\infty} e^{-(1-2j)t} dt = \frac{e^{-(1-2j)t}}{-(1-2j)}\Big|_{0}^{\infty} = \frac{-1}{1-2j}\left(e^{-\infty} - e^{-0}\right) = \frac{1}{1-2j}$

Kararlı

$h(t) = e^{-t}\cos(2t)u(t) \longrightarrow \int_{0}^{\infty} e^{-t}\cos(2t)\,dt < \infty$

Kararlı

$h(t) = t\cos\left(\frac{\pi t}{4}\right)u(t) \longrightarrow \int_{0}^{\infty} t\cos\left(\frac{\pi t}{4}\right) \longrightarrow \infty$

Kararsız

9) $x[n] = u[n-3]$ ve $h[n] = -\delta[n] + \delta[n+2]$ ise $y[n] = ?$

$y[n] = x[n] * h[n] = u[n-3] * \left(-\delta[n] + \delta[n+2]\right) = -\delta[n] * u[n-3] + \delta[n+2] * u[n-3]$

$= -u[n-3] + u[n-1] = \delta[n-1] + \delta[n-2]$

2) Herhangi bir sonlu t aralığında,

- Sonlu sayıda maksimum ve minimum değere sahip olmalıdır.
- Sonlu sayıda süreksizliğe ve bu noktalarda sonlu değere sahip olmalıdır.

⇒ Frekans spektrumları:

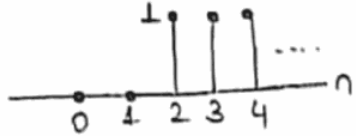
t=-3 t=1

4) $u[n-1]-u[n-2]=?$



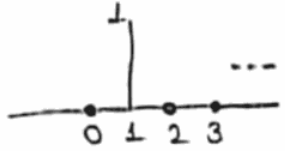
$u[n-1]$

$-$



$u[n-2]$

$=$



$= \delta[n-1]$

5) $u[n-7]-u[n-5]=?$



$u[n-7]$

$-$



$u[n-5]$

$=$



$= -\delta[n-5]-\delta[n-6]$

Ex: $X(f)=\delta(f-f_0)$ ise $x(t)=?$

$$x(t)=\int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df=\int_{-\infty}^{\infty}\delta(f-f_0)e^{j2\pi ft}df=\int_{-\infty}^{\infty}\delta(f-f_0)e^{j2\pi f_0 t}df=e^{j2\pi f_0 t}\underbrace{\int_{-\infty}^{\infty}\delta(f-f_0)df}_{=1}=e^{j2\pi f_0 t}$$

Ex: $X(f)=\delta(f)$ ise $x(t)=?$

$$x(t)=\int_{-\infty}^{\infty}X(f)e^{j2\pi ft}df=\int_{-\infty}^{\infty}\delta(f)e^{j2\pi ft}df=\int_{-\infty}^{\infty}\delta(f)e^{j2\pi\cdot 0\cdot t}df=\int_{-\infty}^{\infty}\delta(f)df=1$$

$$\delta(t)\overset{\mathcal{FF}}{\longleftrightarrow}1$$

$$1\overset{\mathcal{FF}}{\longleftrightarrow}\delta(f)$$

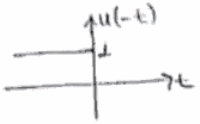
$$e^{j2\pi f_0 t}\overset{\mathcal{FF}}{\longleftrightarrow}\delta(f-f_0)$$

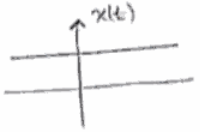
$$e^{-j2\pi f_0 t}\overset{\mathcal{FF}}{\longleftrightarrow}\delta(f+f_0)$$

1) $\sqrt{2}\,e^{j9\pi/4} = \sqrt{2}\left(\cos 9\pi/4 + j\sin 9\pi/4\right) = \sqrt{2}\left(\cos \pi/4 + j\sin \pi/4\right) = \sqrt{2}\left(\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}\right) = 1+j$

2) $x(t) = u(t) + u(-t)$

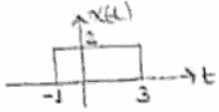






$T \to \infty$
Periyodik değildir

3) $x(t) = 2,\ -1 < t < 3$ ise $x(1-2t) + x(2+t) = ?$

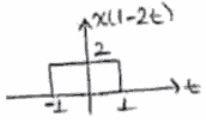


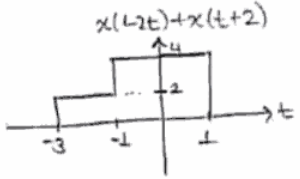
$x(1-2t) \longrightarrow x(t)\quad t : [-1,3]$

$1-2t : [-1,3]$

$1-2t = -1 \quad t = 1$

$1-2t = 3 \quad t = -1$

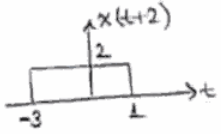




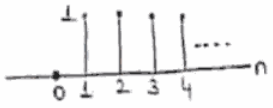
$x(2+t) \longrightarrow x(t)\quad t : [-1,3]$

$2+t : [-1,3]$

$t = -3 \quad t = 1$

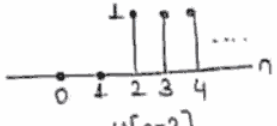


4) $u[n-1] - u[n-2] = ?$



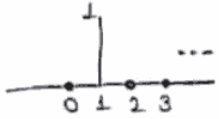
$u[n-1]$

$-$



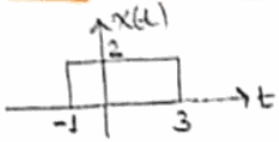
$u[n-2]$

$=$



$= \delta[n-1]$

3) $x(t)=2,\ -1<t<3$ ise $x(1-2t)+x(2+t)=?$

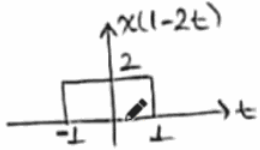


$x(1-2t) \longrightarrow x(t)\quad t:[-1,3]$

$1-2t:[-1,3]$

$1-2t=-1 \quad t=1$

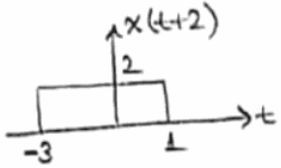
$1-2t=3 \quad t=-1$

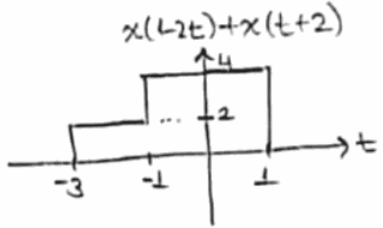


$x(2+t) \longrightarrow x(t)\quad t:[-1,3]$

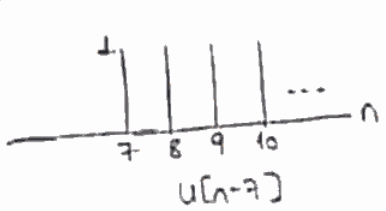
$2+t:[-1,3]$

$t=-3 \quad t=1$



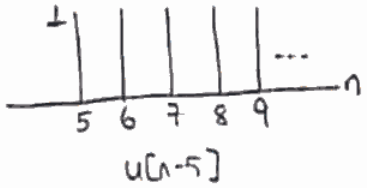


5) $u[n-7]-u[n-5]=?$



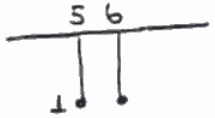
$u[n-7]$

$-$



$u[n-5]$

$=$



$= -\delta[n-5]-\delta[n-6]$

6) $\cos\left(\frac{\pi}{3}n\right)\delta[n+4]=?$

Reminder: $x[n]\,\delta[n]=x[0]\,\delta[n]$

$x[n]\,\delta[n-n_0]=x[n_0]\,\delta[n-n_0]$

$\cos\left(\frac{\pi}{3}n\right)\delta[n+4]=\cos\left(-\frac{4\pi}{3}\right)\delta[n+4]=\cos\left(\frac{4\pi}{3}\right)\delta[n+4]$

CamScanner